

**Tribhuvan University**  
**Institute of Science and Technology**  
**2068**

Bachelor Level/ First Year/ First Semester/ Science  
**Computer Science and Information Technology (CSc. 102)**  
(Fundamental of computer programming)

Full Marks: 60  
Pass Marks: 24  
Time: 3 hours.

Candidates are required to give their answers in their own words as far as practicable.  
The figures in the margin indicate full marks.

**Attempt all questions:**

1. Write an algorithm and flow chart to determine whether a given integer is odd or even and explain it.
2. How can you declare the variable in C? Explain with example.
3. Write a program to find the factorial of a given integer.
4. Explain switch statement with example.
5. Write a program to find the largest and smallest among the given elements in an array.
6. Explain the user-defined functions and its types with example.
7. Write a program to accept two numbers and sort them with using pointer.
8. Explain the passing structure to function with example.
9. Write a program to accept any number and print the sum of that number up to a single digit through recursive function.
10. Explain the pointer to structure with example.

OR

Write short notes on :

- a) Dynamic memory allocation
- b) Opening and Closing file